SIDE 1

STAT 251 · EXAM 2 — Calculation Sheet

A Which procedure? | 1–4 Module 4 | 5–7 Tables & SE | 8–10 CI (side 2) | 11–13 Tests (side 2) | 14 Worked (side 2) | 15–16 MC (side 2)

A · If the question says…	Parameter → estimate	Two-sided CI (estimate ± crit × SE)	Test statistic (under H_0)	Table · df	Conditions (✓ both groups if 2)
"% / proportion of all…", one yes/no group	$p \to \hat{p} = \frac{x}{n}$	$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$ ← uses p_0!	**z**	random · 10% · CI: $n\hat{p}, n\hat{q} \geq 10$; test: $np_0, nq_0 \geq 10$
compare % of two separate groups	$p_1 - p_2 \to \hat{p}_1 - \hat{p}_2$	$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$	$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}$ (lecture: unpooled)	**z**	+ independent groups; S/F for each group
"average / mean of all…", one group, s given	$\mu \to \bar{y}\ (= \bar{x})$	$\bar{y} \pm t^*_{n-1} \frac{s}{\sqrt{n}}$	$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}}$	**t**, $df = n - 1$	random · 10% · Normal-ish or $n > 30$ (heavy skew: $n > 100$)
compare means of two separate groups	$\mu_1 - \mu_2 \to \bar{y}_1 - \bar{y}_2$	$(\bar{y}_1 - \bar{y}_2) \pm t^*_\nu \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$t = \frac{(\bar{y}_1 - \bar{y}_2) - \Delta_0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$, Δ_0 usually 0	**t**, Welch ν (§7) — round **down**	+ independent groups
same subjects before/after, matched pairs	$\mu_d \to \bar{d}$ (one list d_i)	$\bar{d} \pm t^*_{n-1} \frac{s_d}{\sqrt{n}}$, n = # pairs	$t = \frac{\bar{d} - d_0}{s_d/\sqrt{n}}$, d_0 usually 0	**t**, $df = n - 1$	random pairs · d's Normal-ish
Normal X: P(value…) with μ, σ known	—	reverse: $x = \mu + z\sigma$	$z = \frac{x - \mu}{\sigma}$	**z**	told Normal
P(sample mean …) with μ, σ known	—	$\bar{Y} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$	$z = \frac{\bar{y} - \mu}{\sigma/\sqrt{n}}$	**z**	Normal pop. or large n (CLT)
"how many to sample for margin E?"	proportion	$n = \left\lceil \frac{(z^*)^2 \hat{p}(1-\hat{p})}{E^2} \right\rceil$ — no prior estimate → $\hat{p} = .5$; **always round UP**; E as a decimal (3% = .03)		**z***	see §10 table

1 Continuous RVs: PDF / CDF toolkit

$$f(x) \geq 0 \qquad \int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad P(X = x) = 0$$

$$P(a < X < b) = \int_a^b f(x)\,dx = F(b) - F(a)$$

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt \qquad f(x) = F'(x)$$

$P(X > x) = 1 - F(x)$ $<$ and $\leq$ give the **same** answer

$$\mu = E[X] = \int x\,f(x)\,dx \qquad E[X^2] = \int x^2 f(x)\,dx$$

$$\sigma^2 = \mathrm{Var}(X) = E[X^2] - \mu^2 = \int (x-\mu)^2 f(x)\,dx, \qquad \sigma = \sqrt{\sigma^2}$$

Typical asks → do this:

- **find** k: set $\int_{\text{support}} f(x)\,dx = 1$, solve.
- **CDF from PDF**: $F(x) = \int_{\text{low}}^{x} f(t)dt$ on the support; $F = 0$ below, $F = 1$ above (write it piecewise).
- **median** m: solve $F(m) = 0.5$. **p-th percentile**: $F(x_p) = p$.
- "at most / below x" → $F(x)$; "more than" → $1 - F(x)$; "between" → $F(b) - F(a)$.

Integrals: $\int x^n dx = \frac{x^{n+1}}{n+1}$, $\int e^{-\lambda x} dx = -\frac{1}{\lambda} e^{-\lambda x}$.

Practice template $f(x) = kx,\ 0 \leq x \leq 2$: $\int_0^2 kx\,dx = 2k = 1 \Rightarrow k = \frac{1}{2}$. $F(x) = \frac{x^2}{4}$. $P(X < 1) = \frac{1}{4}$. $E[X] = \int_0^2 \frac{x^2}{2} dx = \frac{4}{3}$, $E[X^2] = \int_0^2 \frac{x^3}{2} dx = 2$, $\mathrm{Var} = 2 - \frac{16}{9} = \frac{2}{9}$. Median: $\frac{m^2}{4} = .5 \Rightarrow m = \sqrt{2} \approx 1.414$.

2 Uniform $X \sim U(a, b)$

If you see: "equally likely anywhere between a and b", flat density.

$$f(x) = \frac{1}{b-a},\ a \leq x \leq b \qquad F(x) = \frac{x-a}{b-a}$$

$$P(c < X < d) = \frac{d-c}{b-a} \qquad E[X] = \frac{a+b}{2} \qquad \mathrm{Var} = \frac{(b-a)^2}{12}$$

SOCCER drill starts $U(0, 20)$ min: $P(5 < X < 10) = \frac{5}{20} = .25$; mean 10; percentile: $x_p = a + p(b-a)$.

3 Exponential $X \sim \mathrm{Exp}(\lambda)$

If you see: waiting time until the *first/next* event; λ = **rate per time unit**.

$$f(x) = \lambda e^{-\lambda x},\ x \geq 0 \qquad F(x) = 1 - e^{-\lambda x}$$

$$P(X > x) = e^{-\lambda x}$$

$$P(a < X < b) = e^{-\lambda a} - e^{-\lambda b} \qquad E[X] = \sigma = \frac{1}{\lambda},\ \mathrm{Var} = \frac{1}{\lambda^2}$$

$$\text{percentile: } x_p = \frac{-\ln(1-p)}{\lambda} \qquad \text{median} = \frac{\ln 2}{\lambda}$$

$$\text{memoryless: } P(X > s + t \mid X > s) = P(X > t)$$

LECTURE $\lambda = 5$/hour, first customer between 15 and 60 min. **Convert units first**: 0.25 h, 1 h. $P(0.25 < X < 1) = e^{-5(0.25)} - e^{-5(1)} = .2865 - .0067 \approx$ **.2798**. Rate 5/hour ≠ wait of 5 hours. 2 goals/hour → mean wait $\frac{1}{2}$ h = 30 min.

4 Normal $X \sim N(\mu, \sigma^2)$

If you see: bell-shaped / Normal, μ and SD given.

Trap: $N(\mu, \sigma^2)$ — 2nd number is the **variance**; $\sigma = \sqrt{\sigma^2}$.

standardize: $z = \frac{x - \mu}{\sigma}$ un-standardize: $x = \mu + z\sigma$

Probability: $x \to z \to$ table area (§5). **Find a value/percentile:** area → find it *inside* the table → z → $x = \mu + z\sigma$ (top 10% → left area .90 → $z = 1.28$).

68–95–99.7: within $1\sigma, 2\sigma, 3\sigma$. SOCCER reaction $N(.40, .04^2)$: 68% in .36–.44 s; 95% in .32–.48 s (approx. only — use the table for exact).

5 z-table: area to the LEFT of z

table entry $= \Phi(z) = P(Z \leq z)$

Row = z to 1 decimal, column = 2nd decimal. $z = 1.96$: row 1.9, col .06 → .9750. Negative z → negative half of the table.

Want	Do	e.g. $z = 1.20$ ($\Phi = .8849$)
left $P(Z < z)$	table	.8849
right $P(Z > z)$	$1 - \Phi(z)$	.1151
between a, b	$\Phi(b) - \Phi(a)$	bigger − smaller
outside $\pm\lvert z\rvert$	$2[1 - \Phi(\lvert z\rvert)]$	$z = 2$: $2(.0228) = .0456$
z from an area	find area inside table	.8849 → 1.20

Critical values z^* (memorize / check in table)

Confidence C	80%	90%	95%	98%	99%
look up left area	.9000	.9500	.9750	.9900	.9950
z^* (two-sided)	1.282	1.645	**1.960**	2.326	2.576

One-sided (α in one tail): $\alpha = .10 \to 1.282$, $.05 \to 1.645$, $.01 \to 2.326$. Rule: two-sided C → left area $1 - \frac{\alpha}{2}$.

6 t-table: critical t* for an UPPER tail

Rows = **df**, columns = upper-tail probability $P(T > t^*)$. It does **not** give cumulative areas. Means use t because σ is unknown (we use s) → heavier tails; large df → $t \approx z$.

Pick the column: two-sided CI at C → column $\alpha/2$ (95% → .025; 90% → .05; 99% → .005). One-sided bound/test → column α.

Pick the row: one mean / paired: $n - 1$. Two means: Welch ν, **round DOWN** (e.g. 22.38 → 22). Row missing (e.g. 35 when the table jumps 30→40)? Use the **lower** df row (conservative).

p-value by bracketing: in your df row find the two critical values around $|t|$ → p is between their column headings (one-tail). **Double both** for H_a :≠. Bigger than every entry → p < smallest heading. e.g. $df = 35$: $2.030 < \mathbf{2.40} < 2.438$ → $.01 < p < .025$ (one-tail).

7 Sampling distributions, SE & Welch df

$$\bar{Y} \approx N\left(\mu, \frac{\sigma^2}{n}\right),\ SE(\bar{y}) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

$$\hat{p} \approx N\left(p, \frac{p(1-p)}{n}\right),\ SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

SD = spread of individuals; **SE** = spread of a statistic over repeated samples. n ×4 → SE ÷2.

CLT probability: $P(\bar{Y} > c) = 1 - \Phi\left(\frac{c-\mu}{\sigma/\sqrt{n}}\right)$. Practice: $\mu = .40, \sigma = .04, n = 16$: SE = .01; $P(\bar{Y} > .42) = 1 - \Phi(2.00) = .0228$.

Sample SD (e.g. of paired d_i): $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$

$$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \text{ ("crazy df")}$$

Compute $A = s_1^2/n_1$, $B = s_2^2/n_2$ first: $\nu = (A + B)^2 / \left(\frac{A^2}{n_1-1} + \frac{B^2}{n_2-1}\right)$. If allowed, conservative shortcut $df = \min(n_1, n_2) - 1$.

+ Reverse z lookups (percentiles & cutoffs)

Want	left area to find	z
top 10% / 90th pct	.9000	1.28
top 5% / 95th pct	.9500	1.645
top 2.5%	.9750	1.96
top 1%	.9900	2.33
bottom 10% / 5%	.1000 / .0500	−1.28 / −1.645
middle 90% / 95% / 99%	tails .05 / .025 / .005	±1.645 / ±1.96 / ±2.576
quartiles Q1 / Q3	.2500 / .7500	−0.67 / 0.67

Then $x = \mu + z\sigma$ (individual) or $\bar{y} = \mu + z\,\sigma/\sqrt{n}$ (sample mean). Area between two closest table entries → pick the closer one (or average).

SIDE 2 Inference: CIs · Tests · Worked Examples · MC

8–10 Confidence intervals | 11–13 Hypothesis tests | 14 Worked examples | 15–16 MC must-knows

8 CI master rule

$$\text{CI} = \text{PE} \pm \underbrace{c^* \times SE}_{\text{margin of error } E}$$

1. Parameter? → PE + SE formula (table A, side 1).
2. Critical value: proportions z^* (§5), means t^* (§6).
3. $E = c^* \cdot SE$; interval $(\text{PE} - E,\ \text{PE} + E)$.
4. Conditions (§13) + interpretation sentence.

Interpretation template: "We are 95% confident that the true *[parameter in context]* is between ___ and ___." **Never:** "95% probability the parameter is in this interval." Long-run meaning: 95% of intervals built this way capture the parameter.

Reading a given CI (L, U): $\text{PE} = \frac{L+U}{2}$, $E = \frac{U-L}{2}$, $SE = E/c^*$, width $= 2E$.
Width ↑ when confidence ↑ or n ↓ or variability ↑. Width $\propto 1/\sqrt{n}$ (×4 n → half width).

One-sided bounds ($\alpha = 1 - C$ in ONE tail)

lower: $(\text{PE} - c^*SE,\ \infty)$ upper: $(-\infty,\ \text{PE} + c^*SE)$

c^*: z with left area C (90% → 1.282), or t from column α (90% → .10 column). Professor's equivalent: negative z with left area α, compute PE + z·SE.

Difference CIs: look at 0

Contains 0 → "no difference" plausible · all > 0 → group 1 larger · all < 0 → group 1 smaller.

9 CI — the five formulas, written out

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad \bar{y} \pm t^*_{n-1} \frac{s}{\sqrt{n}} \qquad \bar{d} \pm t^*_{n-1} \frac{s_d}{\sqrt{n}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$(\bar{y}_1 - \bar{y}_2) \pm t^*_\nu \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

t^*_{n-1}: the subscript is the df, not "t* minus 1". Paired: n = number of pairs; compute each d_i in the stated direction first.

10 Sample size for a proportion

$$E = z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \implies n = \left\lceil \frac{(z^*)^2 \hat{p}(1-\hat{p})}{E^2} \right\rceil$$

Use a prior $\hat{p}$ if given, else .5 ($.5 \times .5 = .25$ is the max → safest). **Round UP**, even 9603.1 → 9604.

E ($\hat{p} = .5$)	.01	.02	.03	.04	.05
90% (1.645)	6766	1692	752	423	271
95% (1.96)	**9604**	2401	1068	601	385
99% (2.576)	16590	4148	1844	1037	664

Halve E → 4× the sample. $\frac{1.96^2(.5)(.5)}{.01^2} = 9604$.

11 Hypothesis test workflow

1. H_0 (has "=", the claim/default) and H_a ($<$, $>$, $\neq$) — choose before seeing data.
2. Check conditions (§13).
3. Test statistic = $\dfrac{\text{estimate} - \text{null value}}{SE \text{ under } H_0}$ (table A).
4. Tail from H_a → p-value from the right table.
5. $p \leq \alpha$ → **reject** H_0; $p > \alpha$ → fail to reject H_0.
6. Conclusion in context.

Wording → H_a: "less / decreased / lower" → $<$ · "more / increased / slower(time ↑)" → $>$ · "different / changed / not equal" → $\neq$.
Conclusion template: "Since p = ___ (< or >) α = ___, we (reject / fail to reject) H_0. There is (enough / not enough) evidence that *[H_a in context]*."

12 p-value: tail × table

H_a	z-test (table = left area)	t-test (bracket in df row)
$<$ left	$p = \Phi(z)$	one-tail area beyond $\lvert t \rvert$
$>$ right	$p = 1 - \Phi(z)$	one-tail area beyond $\lvert t \rvert$
$\neq$ two	$p = 2[1 - \Phi(\lvert z \rvert)]$	2 × one-tail bracket

e.g. $z = -1.20$: H_a: $<$ → .1151; H_a: $>$ → .8849; H_a: $\neq$ → .2302.

Rejection regions (critical-value method): $\alpha = .05$ — left: $z < -1.645$ · right: $z > 1.645$ · two: $|z| > 1.96$. For t: same idea with t^* from the df row (column α one-sided, $\alpha/2$ two-sided).

CI ↔ test: two-sided test at α rejects $H_0 : \theta = \theta_0$ exactly when θ_0 is outside the $(1-\alpha)$ CI.

13 Conditions (write them!)

	Random	10%	Shape / counts	Indep. groups
1 prop	✓	✓	CI $n\hat{p}, n\hat{q} \geq 10$; test $np_0, nq_0 \geq 10$	—
2 props	both	both	S/F for both	✓
1 mean	✓	✓	Normal-ish; skew → $n > 30$; heavy skew → $n > 100$	—
2 means	both	both	same, both groups	✓
paired	✓	✓	differences Normal-ish	pairs dependent!

10%: $n \leq 0.10N$ (e.g. $n = 36$ needs population ≥ 360).

14 Worked examples

LECTURE One proportion test (voters): $H_0 : p = .52$, $H_a : p < .52$, $n = 10{,}000$, $\hat{p} = .514$.

$$z = \frac{.514 - .52}{\sqrt{.52(.48)/10000}} = \frac{-.006}{.004996} \approx -1.20$$

left tail: $p = \Phi(-1.20) = .1151$

$.1151 > .05$ → fail to reject; not enough evidence support is below 52%. CI version (uses $\hat{p}$): $.514 \pm 1.96(.004998)$ → (50.42%, 52.38%).

LECTURE Two proportions (cats): F 13/45 = .2889, M 16/74 = .2162, $H_a : p_F - p_M > 0$, unpooled SE:

$$z = \frac{.2889 - .2162}{\sqrt{\frac{.2889(.7111)}{45} + \frac{.2162(.7838)}{74}}} = \frac{.0727}{.0828} \approx 0.878$$

right tail: $p = 1 - \Phi(0.88) = 1 - .8106 = .1894 > .10$ → fail to reject (insufficient evidence).

LECTURE One mean (HS GPA): $H_0 : \mu = 4.1$, $H_a : \mu \neq 4.1$, $n = 114$, $\bar{y} = 4.2317$, $s = .3664$:

$$t = \frac{4.2317 - 4.1}{.3664/\sqrt{114}} = \frac{.1317}{.03432} \approx 3.84, \quad df = 113$$

Row 113 missing → use lower row (e.g. 100). t beyond the last column (one-tail .001) → two-sided $p < .002$ (lecture: $p < .001$; exact .0002). $\alpha = .02$ → reject: mean differs from 4.1.

LECTURE Two means (1st-gen vs not): $n_1 = 17, \bar{y}_1 = 4.1467, s_1 = .3495$; $n_2 = 102, \bar{y}_2 = 4.2466, s_2 = .369$; $H_a : \mu_1 - \mu_2 < 0$.

$$t = \frac{4.1467 - 4.2466}{\sqrt{\frac{.3495^2}{17} + \frac{.369^2}{102}}} = \frac{-.0999}{.0923} \approx -1.083$$

$\nu \approx 22.38 \to 22$, left tail: $p \approx .145$

$.145 > .03$ → fail to reject.

SOCCER One-mean t-test: claim $\mu = .400$ s, suspect slower: $H_a : \mu > .400$; $n = 36$, $\bar{y} = .424$, $s = .060$, $\alpha = .05$.

$$t = \frac{.424 - .400}{.060/\sqrt{36}} = \frac{.024}{.010} = 2.40, \ df = 35$$

Row 35: $2.030 < 2.40 < 2.438$ → $.01 < p < .025 < .05$ → reject; mean reaction time exceeds .400 s. (Row 30 instead: $2.042 < 2.40 < 2.457$, same result.)

SOCCER One-proportion CI: 62 of 100 prefer morning training, 95%:

$$SE = \sqrt{\frac{.62(.38)}{100}} = .04854, \quad E = 1.96(.04854) = .0951$$

$$.62 \pm .0951 = (.5249,\ .7151)$$

S/F: 62, 38 ≥ 10 ✓. Testing $H_0 : p = .50$ instead → $SE = \sqrt{.5(.5)/100} = .05$, $z = \frac{.62 - .50}{.05} = 2.40$.

SOCCER Paired recipe: each keeper's d_i = after − before → $\bar{d}$, s_d (§7 formula) → $t = \dfrac{\bar{d} - 0}{s_d/\sqrt{n}}$, df = pairs − 1. Never use the two-sample formula on paired data.

15 MC must-knows

- **Parameter** (p, μ; population, unknown) vs **statistic** ($\hat{p}, \bar{y}, s$; sample).
- **p-value** = P(data this extreme or more | H_0 true). It is NOT P(H_0 true).
- Fail to reject ≠ "accept H_0" / ≠ proof H_0 is true.
- **Type I**: reject a true H_0 (false alarm), prob $= \alpha$. **Type II**: fail to reject a false H_0 (miss), prob $= \beta$. **Power** $= 1 - \beta$ = P(correctly reject false H_0).
- Power ↑ with larger n, larger effect size, larger α (but Type I risk ↑). ↓α → ↑β.
- Confidence level ↑ → wider CI; n ↑ → narrower CI.
- t vs z: t heavier tails, depends on df; df → ∞ gives z.
- CI uses $\hat{p}$ in SE; test uses p_0 in SE.
- Continuous: $P(X = x) = 0$; probability = area; density can exceed 1 (area can't).
- Uniform flat · Exponential = time until first event (constant rate) · Normal symmetric bell.
- H_a is chosen before seeing data; tail comes only from H_a.

16 Trap list

- p (proportion) ≠ p-value · Var vs SD in $N(\mu, \sigma^2)$ · minutes vs hours for λ.
- z-table is LEFT area; t-table is UPPER-tail critical values — don't read t like z.
- Two-sided: split α (.025 each tail) for CIs; double one-tail p for ≠ tests.
- Paired ≠ two independent groups. Means → t, proportions → z.
- Sample size: round **up**; E as decimal.
- Zero in a difference CI = no difference plausible.

Exam-day decision flow (calc questions)

1 · What is asked? probability of a continuous value → **Module 4** (§1–4) · a range for a parameter → **CI** (§8–10) · evidence about a claim → **test** (§11–12) · "how many to sample" → §10.
2 · Which parameter? $p \cdot p_1 - p_2 \cdot \mu \cdot \mu_1 - \mu_2 \cdot \mu_d$ (table A, side 1).
3 · z or t? proportions & Normal-with-known-σ → **z**; means with s → **t** (df!).
4 · Formula → write it, plug in, SE first.
5 · Tail from H_a (< left, > right, ≠ both) — tests only.
6 · Table: z → left area; t → df row + upper-tail column.
7 · Decide & conclude in context ("fail to reject", "95% confident").

Calculator / table habits

- Keep 4 decimals until the end; round z to 2 decimals to read the table.
- Write the **formula first**, then plug in — partial credit.
- Sketch the curve and shade the area you need every time.
- Keep subtraction order (group 1 − group 2) the same in H_a, estimate and conclusion.